Model paradigm / Architecture	p = 8		p = 10		p = 12		Mean
BITWISE MARGINAL PRIOR (PROPOSED)							
Echo-Prior-Flat	6.0 →	16.83 $^{+1.59}_{-1.62}$	6.3 →	**24.00** $^{+1.93}_{-1.99}$	6.7 →	**29.60** $^{+2.27}_{-2.26}$	**23.48%**
Echo-Prior-End	5.2 →	13.90 $^{+1.35}_{-1.28}$	5.2 →	18.21 $^{+1.54}_{-1.52}$	5.3 →	23.72 $^{+1.81}_{-1.70}$	18.61%
Echo-Prior-Alt	3.5 →	9.09 $^{+0.90}_{-0.85}$	3.3 →	14.16 $^{+1.26}_{-1.26}$	3.3 →	19.27 $^{+1.59}_{-1.61}$	14.18%
DIRECT 2P-ANGLE REGRESSION — GNN BACKBONE							
GNN-TQA	9.3 →	**17.33** $^{+0.60}_{-0.60}$	10.4 →	22.45 $^{+0.70}_{-0.71}$	10.0 →	28.28 $^{+0.94}_{-0.97}$	22.68%
GNN-Flat	0.1 →	4.19 $^{+0.69}_{-0.62}$	0.1 →	7.04 $^{+1.00}_{-0.96}$	0.1 →	10.06 $^{+1.33}_{-1.22}$	7.10%
GNN-End	0.1 →	4.86 $^{+0.71}_{-0.66}$	0.1 →	7.19 $^{+0.92}_{-0.89}$	0.1 →	10.86 $^{+1.24}_{-1.20}$	7.64%
GNN-Alt	0.1 →	4.97 $^{+0.71}_{-0.67}$	0.1 →	7.66 $^{+0.97}_{-0.94}$	0.1 →	11.08 $^{+1.23}_{-1.17}$	7.90%
GNN-RI	0.0 →	7.99 $^{+0.30}_{-0.27}$	0.1 →	9.61 $^{+0.63}_{-0.60}$	0.0 →	14.25 $^{+0.61}_{-0.60}$	10.62%
GNN-INTERP	0.1 →	4.47 $^{+0.60}_{-0.57}$	0.1 →	7.12 $^{+0.86}_{-0.83}$	0.1 →	9.73 $^{+1.00}_{-0.96}$	7.10%
DIRECT 2P-ANGLE REGRESSION — MLP BACKBONE							
MLP-TQA	8.6 →	17.24 $^{+0.61}_{-0.62}$	9.1 →	22.17 $^{+0.68}_{-0.67}$	10.4 →	28.25 $^{+0.93}_{-0.95}$	22.55%
MLP-Flat	0.1 →	4.52 $^{+0.71}_{-0.67}$	0.1 →	7.98 $^{+1.07}_{-1.01}$	0.1 →	11.73 $^{+1.37}_{-1.30}$	8.08%
MLP-End	0.1 →	4.66 $^{+0.68}_{-0.66}$	0.1 →	8.07 $^{+1.03}_{-0.93}$	0.1 →	10.93 $^{+1.29}_{-1.22}$	7.89%
MLP-Alt	0.1 →	5.37 $^{+0.72}_{-0.68}$	0.1 →	7.83 $^{+1.02}_{-0.97}$	0.1 →	10.31 $^{+1.27}_{-1.25}$	7.83%
MLP-RI	0.2 →	6.87 $^{+0.47}_{-0.44}$	0.2 →	9.84 $^{+0.66}_{-0.60}$	0.1 →	10.96 $^{+0.77}_{-0.76}$	9.22%
MLP-INTERP	0.1 →	3.88 $^{+0.56}_{-0.53}$	0.1 →	6.57 $^{+0.74}_{-0.76}$	0.1 →	9.59 $^{+0.99}_{-0.96}$	6.68%

Values report initial → energy-refined ground-state probability (%) on 1,024 held-out MWIS test instances (n = 10, seed = 0; 95% bootstrap CI).